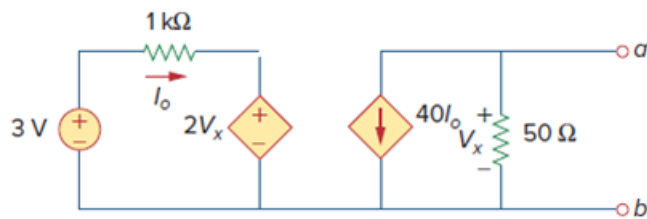


Problem

Find the Thevenin equivalent between terminals a - b of the circuit in Fig. 4.120.

Figure. 4.120:



Step-by-step solution

Step 1 of 8

Refer to Figure 4.120 in the textbook.

Re-draw the circuit diagram by notating with node voltage and loop current.

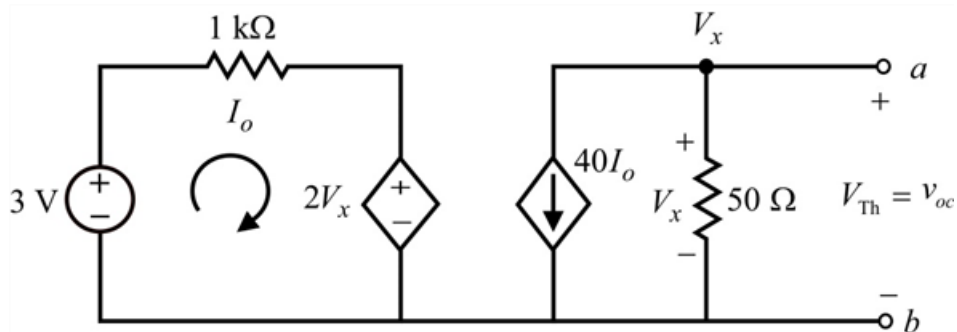


Figure 1

Step 2 of 8

From Figure 1, the open circuit voltage (Thevenin voltage) is,

$$\begin{aligned} V_{Th} &= v_{oc} \\ &= V_x \end{aligned}$$

Apply Kirchhoff's voltage law to the left side of network in Figure 1.

$$-3 + 1000I_o + 2V_x = 0$$

$$1000I_o + 2V_x = 3 \quad \dots\dots (1)$$

Apply Kirchhoff's current law at node V_x .

$$40I_o + \frac{V_x}{50} = 0$$

$$40I_o = -\frac{V_x}{50}$$

$$I_o = -\frac{V_x}{2000} \quad \dots\dots (2)$$

Step 3 of 8

Substitute $-\frac{V_x}{2000}$ for I_o in equation (1) and find the value of V_x .

$$1000\left(-\frac{V_x}{2000}\right) + 2V_x = 3$$

$$-0.5V_x + 2V_x = 3$$

$$1.5V_x = 3$$

$$V_x = 2 \text{ V}$$

The Thevenin voltage across the open circuit terminals a and b is,

$$\begin{aligned} V_{Th} &= V_x \\ &= 2 \text{ V} \end{aligned}$$

Step 4 of 8

Since the circuit has dependent sources, use the test source method to find the Thevenin resistance. Place a unit voltage test source across the open terminals “ a ” and “ b ”, and replace the independent voltage source with a short circuit. And then find the current delivered by the voltage source.

Draw the modified circuit diagram.

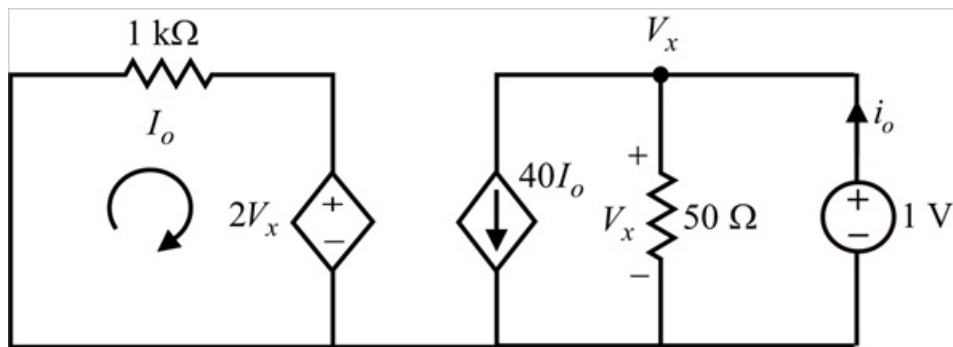


Figure 2

Step 5 of 8

From Figure 2, it is clear that, the voltage V_x is,

$$V_x = 1 \text{ V}$$

Apply Kirchhoff's voltage law to the left side of network in Figure 1.

$$1000I_o + 2V_x = 0$$

$$1000I_o = -2(1)$$

$$I_o = \frac{-2}{1000}$$

$$I_o = -2 \text{ mA}$$

Step 6 of 8

Apply Kirchhoff's current law at node V_x .

$$40I_o + \frac{V_x}{50} - i_o = 0$$

$$40(-2 \times 10^{-3}) + \frac{1}{50} = i_o$$

$$i_o = -0.08 + 0.02$$

$$i_o = -0.06 \text{ A}$$

Step 7 of 8

Determine the equivalent resistance.

$$R_{\text{Th}} = \frac{1 \text{ V}}{i_o}$$

$$= \frac{1}{-0.06}$$

$$= -16.67 \Omega$$

Therefore, the Thevenin resistance is $R_{\text{Th}} = -16.67 \Omega$.

Step 8 of 8

Draw the Thevenin equivalent circuit.

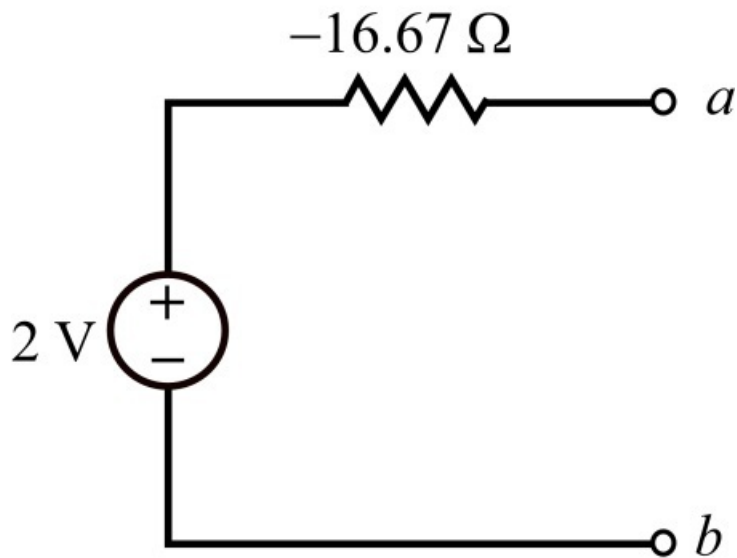


Figure 3